Exact and matheuristic approaches for a real-world process selection and lot-sizing problem in the electrofused grains industry

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Abstract

[TODO: Write last. Skeleton: (1) real-world PSP with co-production in fixed proportions; (2) fifteen years after the original case study, we revisit the problem with a modern MIP solver, quantifying solver progress; (3) we propose relax-and-fix / fix-and-optimize matheuristics and an RF-warm-started monolithic method, and benchmark them against a state-of-the-art hybrid GRASP-ILS with path relinking under a strict equal-resources protocol with pre-registered hypotheses; (4) headline numbers from Sprint 2; (5) public benchmark of 60 instances and a method-choice frontier in the scale × time-budget plane.]

Keywords: Lot sizing, Process selection, Co-production, Matheuristics, Relax-and-fix, Fix-and-optimize, GRASP

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1. Introduction

Electrofused grains, notably silicon carbide and fused aluminum oxide, are synthetic raw materials employed in the manufacture of abrasive, refractory, and ceramic products. Production starts in electric-resistance (Acheson-type)

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or electric-arc furnaces, whose output blocks are subsequently crushed, milled, and classified by sets of screens into a large number of granulometric ranges (grits) with strict physical–chemical specifications. A distinctive feature of the grain-finishing stage is *co-production in fixed proportions*: each configuration of the crushing and classification equipment – hereafter a *production process* – simultaneously yields a specific mix of final products in technologically determined proportions. Planners therefore do not decide production quantities product by product; they decide *which process to operate in each period*, and the product mix follows. Combined with lengthy setups (equipment reconfiguration and cleaning may consume an entire shift), a portfolio of dozens of products, and tight order-book due dates, this makes production scheduling a genuinely hard combinatorial task.

The problem was formalized by Luche et al. (2009) as a *process selection and lot-sizing problem* (PSP): a single-stage, big-bucket model in which at most one process may be assigned to each period and the objective is to minimize the total shortage of the order book (with a small penalty on excess production). At the time, the mixed-integer programming (MIP) model could not be solved to optimality within acceptable times for instances of practical size by the then state-of-the-art solver, which motivated constructive heuristics, local search, and later a GRASP procedure **[TODO: cite thesis]**. Two developments justify revisiting the problem fifteen years later. First, MIP solvers have improved by orders of magnitude in this period through advances in presolve, cut separation, primal heuristics, and parallel tree search (Koch et al., 2022; Achterberg and Wunderling, 2013; Achterberg et al., 2020); whether such progress has effectively closed a practical problem that was intractable in 2011 is an empirical question of direct managerial relevance. Second, MIP-based construction and improvement heuristics – *relax-and-fix* (RF) and *fix-and-optimize* (FO), now standard components of the matheuristics toolbox (Sahling et al., 2009; Helber and Sahling, 2010; Toledo et al., 2015b) – have matured precisely on lot-sizing structures similar to the PSP, but have never been applied to it.

This paper reports a comprehensive computational study of the PSP that confronts three algorithmic paradigms under a strict equal-resources protocol: (i) the monolithic MIP model solved by a modern commercial solver; (ii) temporal-decomposition matheuristics (RF+FO and an RF-warm-started monolithic variant, RF+MIP); and (iii) a state-of-the-art hybrid metaheuristic combining reactive GRASP, iterated local search, variable neighborhood descent, tabu components, and cross-run path relinking (GRASP-ILS-PR). All methods run on the same machine, with the same wall-clock budgets and explicit thread accounting, and the central comparisons were specified as pre-registered hypotheses before the experiments were executed. Rather than asking only *whether* a heuristic beats the solver, we map *when* each method is the tool of choice as a function of instance scale (planning horizons from $T = 19$ to $T \approx 190$ periods) and available computing time (from operational re-planning budgets of 5–10 minutes to tactical budgets of 1–3 hours). **[SPRINT-2 DATA: one-sentence preview of the frontier result once Sprint 2 hypotheses H1–H2 are adjudicated.]**

The contributions are fourfold. First, we quantify fifteen years of MIP solver progress on a documented real-world problem: instances that exhibited large optimality gaps in 2011 are now solved in seconds **[SPRINT-2 DATA: exact figures]**, while a hard subclass remains open at the largest scales – a data point for the solver-progress literature (Koch et al., 2022) grounded in an industrial application. Second, we develop and specify, at a reproducibility level rarely reported (bound-update mechanics, per-window time-limit policies, and budget guards included), RF and FO matheuristics tailored to the co-production structure of the PSP, together with an RF+MIP hybrid that dominates the cold-started solver by construction. Third, we provide what is, to the best of our knowledge, the first publicly available benchmark for lot sizing with co-production in fixed proportions: 60 instances spanning ten problem scales, with best-known solutions, dual bounds, and full per-run trajectories, released in open formats. Fourth, we derive managerial insights for the electrofused grains mill, including the practical consequence that order-book scheduling can now be re-optimized within the minutes-scale budgets of daily re-planning **[SPRINT-2**

DATA: support with H1 outcome].

The remainder of the paper is organized as follows. Section 2 reviews the related literature and positions the PSP within the lot-sizing and scheduling taxonomy. Section 3 describes the industrial setting and the mathematical model. Section 4 presents the RF, FO, and RF+MIP methods, and Section 5 summarizes the hybrid GRASP-ILS-PR baseline. Section 6 reports the experimental protocol and results, and Section 7 concludes.

2. Related work

We review four streams: the lot-sizing and scheduling taxonomy in which the PSP is embedded (§2.1); applications featuring process selection and co-production in fixed proportions (§2.2); MIP-based matheuristics of the relax-and-fix and fix-and-optimize families (§2.3); and the solver-progress and experimental-methodology literature that underpins our protocol (§2.4).

2.1. Lot sizing and scheduling: taxonomy and reviews

Dynamic lot-sizing models that integrate sequencing decisions are classically organized by their time structure. Small-bucket models allow at most one (or a few) setups per period: the discrete lot-sizing and scheduling problem (DLSP; Fleischmann, 1990) imposes all-or-nothing production, the continuous setup lot-sizing problem (CSLP; Karmarkar and Schrage, 1985) relaxes full capacity usage, and the proportional lot-sizing and scheduling problem (PLSP; Drexl and Haase, 1995) allows the setup state to change once within a period. The general lot-sizing and scheduling problem (GLSP; Fleischmann and Meyr, 1997) subsumes these structures through micro-periods, and the capacitated lot-sizing problem with sequence-dependent setups (Haase, 1996) extends the big-bucket tradition. Comprehensive surveys trace the evolution of this family: Drexl and Kimms (1997) and Karimi et al. (2003) for the foundations, Jans and Degraeve (2008) for industrial extensions, Buschkühl et al. (2010) for solution approaches, Glock et al. (2014) at the tertiary level, and, most relevant to our positioning, the classification of simultaneous lot-sizing and scheduling models by Copil et al. (2017), recently extended to secondary resources by Wörbelauer et al. (2019). Within this taxonomy the PSP is a single-stage, single-machine, big-bucket model with at most one "product" – in fact, one *process* – per period; its distinctive trait, absent from the standard classes, is that the assignment decision releases a technologically fixed *vector* of products rather than a single item. Demand is met against cumulative order-book quantities with shortage minimization, which relates the model to service-level-oriented variants rather than to the cost-minimization mainstream (Pochet and Wolsey, 2006).

2.2. Process selection and co-production in fixed proportions

The PSP was introduced in the electrofused grains context by Luche et al. (2009), who combined process selection with single-stage lot sizing, proposed a constructive heuristic, and reported that instances of practical size defeated the MIP technology of the time. A structurally similar coupling of process configuration, lot sizing, and scheduling arises in the molded pulp industry (Martínez et al., 2018), where each mold configuration also determines a mix of items, and in small foundries, where each furnace alloy enables a set of castable parts (de Araujo et al., 2008). Related recipe- or process-based settings include animal nutrition plants (Toso et al., 2009; Clark et al., 2010), chemical plants in which one item may be produced by alternative processes (Cunha and Santos, 2017), and integrated soft-drink production where tank lots feed bottling lines (Ferreira et al., 2009; Toledo et al., 2015a). Co-production in fixed (or stochastic) proportions is also the defining feature of semiconductor planning, where each wafer lot "bins" into multiple performance grades (Leachman, 2002; Ng and Fowler, 2007), and connects to multi-period cutting-stock models in which a cutting pattern plays the role of a process (Silva et al., 2023). Across this literature, exact approaches are typically abandoned at realistic scales in favor of tailored heuristics; none of these works, however, revisits the abandoned monoliths with current solver technology, and public benchmarks for fixed-proportion co-production are, to the best of our knowledge, unavailable.

2.3. Relax-and-fix and fix-and-optimize matheuristics

Relax-and-fix constructs solutions by partitioning the integer variables – most naturally along the time axis – and solving a sequence of MIPs in which one window is kept integral, earlier windows are fixed, and later windows are relaxed (Dillenberger et al., 1994; Pochet and Wolsey, 2006). Fix-and-optimize (also called exchange or MIP-based improvement) iteratively re-optimizes subsets of integer variables while fixing the remainder at incumbent values; it was popularized for multi-level capacitated lot sizing by Sahling et al. (2009) and Helber and Sahling (2010). The combination of the two – RF for construction, FO for improvement – was systematized by Toledo et al. (2015b) on multi-level lot-sizing and glass-container problems, and industrial variants have since accumulated: integrated pulp and paper planning (Santos and Almada-Lobo, 2012), later improved by variable neighborhood search (Figueira et al., 2013) and by structure-exploiting matheuristics (Furlan et al., 2024); beverage plants with temporal cleanings (Toscano et al., 2020); and perishable-product lines, where relax-and-fix also supplies competitive dual bounds (Soler et al., 2021). Iterative MIP-based neighborhood searches for lot sizing and scheduling were explored by James and Almada-Lobo (2011), and the genre is surveyed under the "matheuristics" label by Fischetti and Fischetti (2018). Our RF+FO follows the time-window template of Toledo et al. (2015b); the differentiating elements are the co-production structure (windows re-optimize the process schedule, with all product balances linked through cumulative demand), a fully specified budget-control policy whose implementation details we report for reproducibility, and the RF+MIP variant in which the RF solution warm-starts the *monolithic* model – a hybrid that is, by construction, never worse than the cold-started solver under equal budgets.

2.4. Solver progress and experimental methodology

Machine-independent measurements document speedups of several orders of magnitude in MIP solving between the early 2000s and 2020, driven by presolve, cutting planes, primal heuristics, and parallel search (Koch et al., 2022; Achterberg and Wunderling, 2013; Achterberg et al., 2020; Bixby, 2012). This progress reframes older intractability findings: conclusions drawn against CPLEX 11–12 – as in the original PSP studies – need not hold against CPLEX 22, and revisiting them yields calibrated evidence of practical value. On the methodology side, our experimental protocol adopts community standards for heuristic–exact comparisons: performance profiles (Dolan and Moré, 2002), time-to-target plots for randomized methods (Aiex et al., 2007), and public release of instances and solutions in the spirit of MIPLIB (Gleixner et al., 2021). The metaheuristic baseline itself follows established components: GRASP (Feo and Resende, 1995) with reactive parameter control (Prais and Ribeiro, 2000), iterated local search (Lourenço et al., 2010), and path relinking (Laguna and Martí, 1999; Resende and Ribeiro, 2016). Finally, the managerial framing of short re-planning budgets draws on the rolling-horizon tradition initiated by Baker (1977).

Gap. In summary: (i) the PSP with co-production in fixed proportions has not been revisited since 2011, despite solver progress that plausibly changes its practical status; (ii) matheuristic studies seldom report primal–primal comparisons against a modern solver under strictly equal wall-clock and thread budgets, leaving the "method of choice" question anecdotal; and (iii) no public benchmark exists for this problem class. This paper addresses all three gaps, with hypotheses registered before experimentation.

3. The process selection and lot-sizing problem

3.1. Industrial setting

We study the grain-finishing stage of a manufacturer of electrofused grains (silicon carbide and fused aluminum oxide). Upstream, raw materials are fused in electric furnaces and cooled into blocks of roughly twenty tonnes; the finishing stage crushes, mills, and screens this material into final products defined by granulometric ranges and physical–chemical specifications. The finishing equipment can be arranged in a number of alternative configurations – the *production processes*. Two features of these processes drive the planning problem. First, *co-production in fixed proportions*: operating process j for one period yields a technologically determined quantity a_{ij} of every product i simultaneously; the planner selects processes, not product quantities. Second, *setup magnitude*: reconfiguring and cleaning the equipment between processes consumes a substantial fraction of a planning period, which motivates a big-bucket representation in which at most one process is operated per period. Demand stems from an order book with due dates; the commercial priority is meeting the order book, so the objective minimizes total shortage, with a small penalty discouraging gratuitous excess production (excess is held as inventory and may serve future orders, but ties up capital and storage). **[TODO: One paragraph with anonymized company facts: number of products/processes in the order-book instances, period length, planning horizon – from thesis Ch. 2, updated.]**

3.2. Mathematical model

Let I denote the set of products, J the set of processes, and $T = \{1, \ldots, |T|\}$ the set of periods. Parameter $a_{ij} \geq 0$ gives the output of product i when process j is operated for one period, and $d_{it} \geq 0$ the demand of product i due at period t. The decision variables are $x_{jt} \in \{0, 1\}$, equal to 1 if process j is operated in period t, and the nonnegative continuous variables f_{it} and e_{it}, measuring the shortage and the excess, respectively, of product i with regard to its *cumulative* demand up to period t. The model, referred to as MFP (minimization of shortage of production, after Luche et al., 2009), reads:

$$\min \quad \sum_{i\in I}\sum_{t\in T}\left(f_{it} + \gamma\, e_{it}\right) \tag{1}$$

$$\text{s.t.} \quad \sum_{j\in J}\sum_{\tau\le t} a_{ij}\, x_{j\tau} + f_{it} - e_{it} \;=\; \sum_{\tau\le t} d_{i\tau}, \qquad i\in I,\ t\in T, \tag{2}$$

$$\sum_{j\in J} x_{jt} \;\le\; 1, \qquad t\in T, \tag{3}$$

$$x_{jt}\in\{0,1\}, \quad f_{it},\, e_{it}\ge 0, \qquad i\in I,\ j\in J,\ t\in T. \tag{4}$$

Constraints (2) balance cumulative production against cumulative demand for every product and period, so that f_{it} captures lateness with respect to due dates (not only end-of-horizon shortage), and constraints (3) allow at most one process – possibly none – per period. The weight $\gamma = 0.001$ makes excess lexicographically subordinate to shortage in practice. The model has $|J|\,|T|$ binary and $2|I|\,|T|$ continuous variables; for the largest instances considered here ($|J| = 159$, $|T| = 190$) this amounts to 30,210 binaries. **[TODO: Dimensions table Real–10X (binaries, continuous, constraints) – generate from instance files.]**

Model MFP is a single-stage, big-bucket member of the simultaneous lot-sizing and scheduling family (Copil et al., 2017), with an assignment-type setup structure per period; its distinctive element is that each period's assignment releases the fixed output vector $(a_{ij})_{i\in I}$, coupling all products through constraints (2).

3.3. Instances and empirical difficulty

Our benchmark comprises ten order-book instances built on real company data (set Real, $|T| \in \{19, 25\}$) and horizon-replicated sets obtained by the rule documented in the original study **[TODO: cite thesis §6.2.2]**: the number of periods is multiplied by k and the demand of the original horizon is repeated exactly in each 19-period block, yielding sets 2X–5X ($k \in \{2, \ldots, 5\}$, inherited) and the new sets 8X and 10X introduced in this paper ($k \in \{8, 10\}$, $|T|$ up to 190). Section 6.1 details the generation and its validation against the inherited sets. Demand is therefore periodic by construction across the whole family – a property we state explicitly because it keeps the generation rule constant along the scale axis, avoiding confounds; empirically, periodicity does not make the instances easy for the solver (Section 6).

Two empirical observations shape the experimental design. First, difficulty is heterogeneous within sets: a subclass of instances (e.g., **[TODO: confirm final list: IncT3x_7, IncT4x_7, IncT5x_10]**) exhibits both large solver gaps and weak dual bounds – on IncT5x_10, the best known solution still lies about 17% above the three-hour dual bound, so the true optimality gap is unknown. Second, the solver's parallel search, although deterministic for a fixed budget, is budget-nonmonotone in the primal sense: on three instances the one-hour run returned a better incumbent than the independent three-hour run. Both observations motivate reporting gaps against best-known solutions (BKS), defined as the minimum objective over *all* methods, budgets, and seeds considered in this study, rather than against dual bounds alone.

4. Matheuristics

We propose temporal-decomposition matheuristics that use the MIP solver as a subroutine over restricted versions of model (1)–(4). A *schedule* is a vector $s \in (\{0\} \cup J)^{|T|}$ assigning to each period a process or idleness; any schedule is feasible, and its cost is computed by an evaluator that replicates the MIP objective exactly (the evaluator reproduces the solver objective on reference solutions to within 10^{-2}, a consistency check enforced throughout).

4.1. A single restricted model with bound control

Relax-and-fix requires period-dependent variable regimes (fixed, binary, or linearly relaxed), whereas algebraic modeling systems attach integrality to whole variables. We therefore build, once per instance, a single model with two variable families: binary x_{jt} and continuous $\bar{x}_{jt} \in [0, 1]$, every occurrence of the assignment variable in (2)–(3) being replaced by $x_{jt} + \bar{x}_{jt}$. Period regimes are then governed purely by bounds: a *relaxed* period fixes $x_{jt} = 0$ and frees $\bar{x}_{jt}$; a *binary* period frees x_{jt} and fixes $\bar{x}_{jt} = 0$; a *fixed* period sets x_{jt} to the incumbent value and $\bar{x}_{jt} = 0$. Between consecutive solves only bounds change, so the model is generated once. An implementation detail proved decisive: updating bounds record-by-record dominated the running time, whereas batch record updates reduced the construction phase from hundreds of seconds to below ten on the largest instances. We report this because the practicality of temporal decomposition in algebraic modeling environments hinges on it. **[TODO: cross-check final wording with Codex implementation notes.]**

4.2. Relax-and-fix construction

The construction sweeps time windows of width σ advancing by $\delta < \sigma$ periods (overlap $\sigma - \delta$), the last window anchored at the end of the horizon so that binary windows cover $1..|T|$. At each iteration the current window is binary, all earlier periods are fixed at the incumbent partial schedule (only the first δ periods of the previous window are fixed definitively; the overlap is re-optimized), and all later periods are relaxed. Each window MIP is solved with relative gap tolerance 10^{-3} and a dynamic time limit $\ell = \max\{5,\ \min\{\ell_{\mathrm{RF}},\ B^{\mathrm{rem}}_{\mathrm{RF}}/w^{\mathrm{rem}},\ B^{\mathrm{rem}}\}\}$ seconds, where ℓ_{RF} caps the per-window limit, $B^{\mathrm{rem}}_{\mathrm{RF}}$ and B^{rem} are the remaining construction and global budgets, and w^{rem} the number of windows still to solve. The construction budget is $B_{\mathrm{RF}} = \min\{0.25\,B,\ w\,\ell_{\mathrm{RF}}\}$ for global budget B and w windows. If a window solve yields no integer incumbent, its limit is doubled once; on a second failure – and whenever the global budget guard triggers – all undecided periods are completed by a greedy rule based on the evaluator, so the construction never returns an incomplete schedule. Under very short budgets, where the per-window allowance drops below a threshold ($B_{\mathrm{RF}}/w < 10$ s), windowed construction is skipped altogether in favor of the greedy rule, and the entire budget is transferred to the improvement phase. **[SPRINT-2 DATA: confirm final adaptive-construction threshold and greedy details after Task 3.2.]**

4.3. Fix-and-optimize improvement

Improvement re-optimizes windows of ω consecutive periods: binaries inside the window are freed, all others are fixed at the incumbent, and the window MIP is solved with relative gap tolerance 10^{-2}, per-window limit ℓ_{FO}, and the incumbent supplied as a MIP start (accepted starts were verified in the solver log). A solution is accepted if it improves the global objective by more than 10^{-6}; since the start is feasible for every window subproblem, accepted moves never degrade the incumbent, and the method reports the best solution found in *any* phase of the run. Windows first sweep the horizon with step $\omega/2$; after a full sweep without improvement, contiguous windows with random start positions (seeded) are drawn until the budget is exhausted. Parameters were tuned on the two hardest 5X instances over the grid $\omega \in \{12, 20, 28\} \times \ell_{\mathrm{FO}} \in \{30, 60\}$ (Section 6 discloses this choice and its implications), selecting $\omega = 20$, $\ell_{\mathrm{FO}} =$ 30 s: on $|T| = 95$, $\omega = 12$ proved too myopic (mean gap +7.2% to BKS) while $\omega = 20$ attained −1.1%, i.e., new best-known solutions.

4.4. Warm-started monolith (RF+MIP)

The third method uses relax-and-fix only to produce a high-quality integer solution quickly (on the order of seconds to two minutes), then solves the *monolithic* model with the remaining budget, zero gap tolerance, and the RF solution as a MIP start. Relative to the cold-started solver under the same wall-clock budget, RF+MIP sacrifices a fraction of branch-and-bound time in exchange for a strong initial incumbent; unlike RF+FO, it retains the solver's global view and dual bound. **[SPRINT-2 DATA: summarize when each variant wins, from the frontier map; discuss the IncT10x_5 anomaly with the Task 3.1 post-mortem evidence.]**

4.5. Parameter summary

All runs record full provenance (solver version, threads, hardware, seeds) and per-window logs (window position, wall time, incumbent profile, and post-solve duals of (2) and (3)); the logs are released with the benchmark.

Algorithm 1 RF+FO (sketch; RF+MIP replaces lines 6–9 by one warm-started solve of the full model)

1: build restricted model with bound control (§4.1)
2: **if** $B_{\mathrm{RF}}/w < 10$ s **then** $s \leftarrow$ greedy construction
3: **else** $s \leftarrow$ relax-and-fix sweep (§4.2)
4: **end if**
5: $s^* \leftarrow s$
6: **while** budget remains **do**
7:     choose next window (sweep, then random after a no-improvement sweep)
8:     fix all periods outside the window at s^*; free the window; MIP-start at s^*
9:     solve with limit ℓ_{FO}, tolerance 10^{-2}
10:     **if** improvement $> 10^{-6}$ **then** update s^*
11:     **end if**
12: **end while**
13: **return** best schedule found in any phase

5. Hybrid metaheuristic baseline

To situate the matheuristics against the strongest *pure* metaheuristic we could build – one that never calls a MIP solver – we developed a hybrid procedure (GRASP-ILS-PR) combining components with strong track records in the GRASP literature. We summarize it here and refer to the released source code for full detail. **[TODO: Cross-check every component description against docs/metaheuristicas.docx and grasp_ils_psp.py before submission; confirm elite-pool size, tabu tenure rule, and perturbation-strength schedule.]**

Solutions are schedules $s \in (\{0\} \cup J)^{|T|}$ evaluated by the same exact evaluator used by the matheuristics (Section 4), so all methods optimize literally the same objective. Construction is a reactive GRASP (Feo and Resende, 1995; Prais and Ribeiro, 2000): candidate processes for each period are ranked by my-

Table 1: Matheuristic parameters (final values).

Phase	Parameter	Value	Meaning
RF	σ / δ	10 / 5	window width / step
RF	ℓ_{RF}	120 s	per-window cap
RF	optcr	10^{-3}	window gap tolerance
RF	B_{RF}	$\min\{0.25B,\, w\ell_{\mathrm{RF}}\}$	construction budget
FO	ω / step	20 / 10	window width / sweep step
FO	ℓ_{FO}	30 s	per-window cap
FO	optcr	10^{-2}	window gap tolerance
all	threads	0 (all cores)	explicit solver threads

opic shortage reduction, the restricted candidate list parameter α being drawn from a pool whose probabilities adapt to observed solution quality. Local search is a variable neighborhood descent over process-exchange and Or-opt-style relocation moves on the schedule; for long horizons, moves are restricted by a sliding time window with a tabu mechanism to keep iteration costs bounded. The improvement loop is an iterated local search (Lourenço et al., 2010) whose perturbation applies multi-bridge reinsertions of schedule segments, with adaptive strength. Ten workers run in parallel from different seeds and share an elite pool through cross-run path relinking (Laguna and Martí, 1999; Resende and Ribeiro, 2016), and the incumbent trajectory of every worker is logged.

Under the equal-resources protocol of Section 6, GRASP-ILS-PR occupies a deliberately generous position: its ten workers exploit the full machine for the whole wall-clock budget, and its per-instance result is the best of ten runs. This makes the comparison against the matheuristics and the solver *a fortiori*: any deficit observed for GRASP-ILS-PR cannot be attributed to a resource handicap. Historically, this procedure is the modernized descendant of the heuristics proposed for the PSP in the original studies (Luche et al., 2009) **[TODO: + thesis cite]**, and it improves substantially upon them **[SPRINT-2 DATA: one sentence with v1-to-v2 improvement figures from the master tables]**; its role in this paper, however, is that of a calibrated baseline representing the pure-metaheuristic paradigm.

6. Computational experiments

6.1. Benchmark

[TODO: Draft after Sprint 3 report. Content plan below; benchmark subsection is referenced from Section 3.]

7. Conclusions

[TODO: Draft last. Content plan:]

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Data availability

The complete benchmark (60 instances in open format), all solution files, and the source code of the matheuristics and analysis pipeline are available at **[TODO: Zenodo DOI + GitHub URL after anonymization decision]**.

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